METRIC

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

HEAT EXCHANGER CLEANING COMPOUNDS

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers liquid form heat exchanger cleaning compounds, hereinafter referred to as "compounds," intended for removing scale deposits from heat exchangers and distilling units.

2. CLASSIFICATION.

2.1 Materials. Unless otherwise specified herein, materials shall be in accordance with the manufacturer's material specifications. The compound shall be a uniform liquid or be able to be made uniform with mild mixing per manufacturer's instructions.

3. SALIENT CHARACTERISTICS.

3.1 Hydrogen-ion concentration (pH). The compound shall be an acidic solution such that the solution, when prepared for use according to manufacturer's recommendations, shall have a pH less than 2.2 when determined in accordance with ASTM E70 or ASTM D1293.

3.2 Inorganic acid content. Inorganic acid-based compounds, as prepared for use, shall contain not more than 10 percent by weight hydrochloric acid (HCl, CAS 7647-01-0) or 15 percent by weight orthophosphoric acid (H_3PO_4, CAS 7664-38-2) when determined in accordance with the manufacturer's standard practice.

3.3 Citric acid content. Citric acid based compounds, as prepared for use, shall contain not more than 10 percent by weight citric acid ($C_6H_8O_7$, CAS 77-92-9) when determined in accordance with the manufacturer's standard practice.

3.4 Organic acid content. Organic acid-based compounds or blends from the carboxylic group or others, as prepared for use, shall contain not more than 20 percent by weight organic acids when determined in accordance with the manufacturer's standard practice.

3.5 Acid-replacement and alternative descaling technologies. Acid-replacement, salt-based, or other alternative descaling compounds will be permitted where the compound meets the scale dissolving, material compatibility, corrosion, protective film, safety, and product conformance requirements of this CID. Such compounds shall be identified by the manufacturer's formulation type and recommended use concentration.

3.6 Excluded ingredients. The compound shall not contain sulfuric acid or organic solvents.

3.7 Toxins, carcinogens, and reproductive stressors. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Conformance shall be determined by a health hazard assessment (HHA) in accordance with 3.7.1. Approval will be in effect for a maximum of 3 years, after which, reissue may be requested for the same formulation as the original approval. As specified (see 7.4), manufacturers shall provide a copy of the approval applicable to offered products to the contracting activity before the product will be accepted.

Beneficial comments, recommendations, additions, deletions, clarifications, and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

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3.7.1 HHA. An HHA will be conducted to ensure conformance to 3.7. The Navy and Marine Corps Force Health Protection Command (NMCFHPC) will evaluate the compound using data provided by the manufacturer/distributor to the NMCFHPC. The NMCFHPC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCFHPC to the Government for evaluation. The HHA process is described on the NMCFHPC's website, https://www.med.navy.mil/Navy-and-Marine-Corps-Force-Health-Protection-Command/Environmental-Health/Industrial-Hygiene/Acquisition-Technical-Support/Health-Hazard-Assessments-HHA/.

3.7.2 Safety data sheet (SDS). When specified (see 7.4), the contracting activity shall be provided with an SDS with each contract bid proposal or prior to contract award. The SDS shall be provided in accordance with the requirements of FED-STD-313 and 29 CFR 1910.1200, Hazardous Communication Standard. The SDS shall also be included with each shipment of the compound.

3.8 Scale dissolving ability. The compound, at usage concentration, shall dissolve calcium carbonate scale at a rate of 1 gram of solid calcium carbonate (chalk or clam shell) in 50 milliliters of compound at 25 °C in less than 1 hour.

3.9 Deleterious effects on non-metallic components. The compound shall not cause degradation of the non-metallic components listed in table I. In order to determine the effects of exposure to the candidate compound, the materials listed in table I shall be exposed to the compound at usage concentration for a period of 70 hours at a temperature of 48 °C. The following properties shall be examined.

3.9.1 Hardness change. Durometer or Rockwell hardness tests (ASTM D2240 and ASTM D785, respectively) shall be performed on non-metallic specimens before and after exposure to the candidate compound. Maximum change in hardness shall be ±5 points.

3.9.2 Permanent volume change. The change in volume of the non-metallic specimens following exposure to the compound shall be determined using ASTM D471 with the exception that the volume following exposure shall be determined after the specimen has been allowed to equilibrate in ambient conditions for 24 hours. The change in volume determined in this manner shall not exceed ±3.5 percent.

3.9.3 Other visual effects. No cracking of the surface when viewed through a 7-power magnifying glass shall be permitted.

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TABLE I. Non-metallic materials list.

MATERIAL	SPECIFICATION	TITLE
Synthetic Rubber (Acrylonitrile-butadiene)	MIL-G-21610, type I	Gaskets, Heat Exchanger, Various Cross Section Rings, Synthetic Rubber
Synthetic Rubber (Acrylonitrile-butadiene, NBR, 70-hardness)	MIL-PRF-6855, class 1, grade 70	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes, General Specification For
Fluorocarbon Rubber (FKM, 70-80-hardness)	SAE AMS3216	Elastomer: Fluorocarbon (FKM) Rubber High-Temperature – Fluid Resistant Low Compression Set / 70 to 80 Type A Hardness for Products in Fuel Systems and Specific Engine Oil Systems
Fluorocarbon Rubber (FKM, 75-hardness)	SAE AMS7276	Rubber: Fluorocarbon (FKM) High Temperature – Fluid Resistant Low Compression Set/70 to 80 Type A Hardness for Seals in Fuel Systems and Specific Engine Oil Systems
Tetrafluoroethylene Resin (TFE)	SAE AS8791	Hydraulic and Pneumatic Retainers (Backup Rings), Polytetrafluoroethylene (PTFE) Resin
Nitrile Rubber (Acrylonitrile-butadiene, NBR, 88-hardness)	ASTM D2000, class A, low flexibility	Standard Classification System for Rubber Products in Automotive Applications
Polychloro-trifluoroethylene (PCTFE)	SAE AMS3650	Rods, Sheets, and Molded Shapes, Polychlorotrifluoroethylene (PCTFE), Unplasticized
Nylon 101 (Polyamide Type 6/6)	SAE AMS3617	Polyamide Type 6-6, Plastic Moldings and Extrusions
Delrin AF	Proprietary	Commercial: Polymer Corporation DE 588 AS or equivalent (polyoxymethylene with 20% polytetrafluoroethylene (PTFE))
Garlock 19	Proprietary	Commercial: 80 durometer styrene-butadiene rubber (SBR) sheet reinforced with 10.8-ounce. cotton cloth insert

3.10 Corrosive effect on copper alloys. The compound, at usage concentration, shall have a corrosive effect on copper alloys (UNS C70600, UNS C71500, and UNS C92200) of not more than 20 mils per year when determined using ASTM G31 for a 6-hour test at 48 °C.

3.11 Effect on protective films. Any candidate compound shall not significantly alter the protective film kinetics of copper alloys, i.e., the compound shall not interfere with the formation of the protective oxide layer that naturally develops on copper alloys exposed to clean natural seawater, substitute ocean water in accordance with ASTM D1141, or other artificial seawater, meeting the criteria of table II. In order to quantify/determine the effect(s) of a candidate compound, the following shall be performed.

3.11.1 Specimen preparation. A minimum of seven UNS C71500 (70/30) copper alloy specimens (one unconditioned, three to be cleaned with the candidate compound, and three controls) each with a minimum total surface area of 17.5 square centimeters shall be prepared by polishing all exposed surfaces of the specimen first with 240-grit abrasive paper followed by 400-grit abrasive paper to remove surface films (including existing protective layers) and expose a clean metallic surface. No discoloration of the surface shall be visible following polishing. The specimens shall be rinsed in clean deionized water to remove abrasive and abraded particles.

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3.11.2 Baseline curve. A potentiodynamic polarization curve shall be obtained for one freshly prepared specimen exposed to seawater for the purpose of the test (not a conditioned or control specimen). The seawater shall be from the same source as the exposure tests in 3.11.3. The specimen shall be allowed to come to equilibrium in non-flowing seawater for a minimum of 30 minutes, but no more than 60 minutes, prior to initiation of the scan. An anodic polarization curve shall be generated at a scan rate of 0.6 V/hour from the corrosion potential (Ecorr) to +0.2 V against a saturated calomel electrode for this and all subsequent polarization scans. Tests shall be in accordance with ASTM G61 with the following modifications:

a. Where 3.56 percent sodium chloride solution is called for in ASTM G61, clean natural seawater shall be used.

b. Do not purge the solution to remove oxygen.

c. Do not perform the reverse polarization scan.

3.11.3 Initial seawater exposure. The remaining six 70/30 copper alloy specimens prepared in 3.11.1 shall be exposed to continuously refreshed, once-through (not recirculated) natural seawater in accordance with 3.11.7 for a minimum of 3 weeks (21 days). Following the initial minimum 21-day exposure, potentiodynamic polarization curves shall be obtained for the control specimens as specified in 3.11.2.

3.11.4 Control specimen criteria. The anodic current density over the range of cupronickel potentials shall decrease by a minimum of two orders of magnitude from the specimen tested in 3.11.2. If this criterion is not met for the control specimens, the test shall be repeated starting at 3.11.1.

3.11.5 Test specimens. Following the initial 21-day exposure, three test specimens (not the control specimens in 3.11.4) exposed to natural seawater as described in 3.11.3 shall be cleaned by immersion in the compound at usage concentration for 6 hours with 5 minutes of stirring each half hour. Following cleaning, the test specimens shall be rinsed with water and reimmersed in continuously refreshed, once-through natural seawater for an additional 30 days. After the 30-day seawater exposure, obtain potentiodynamic polarization curves for the test specimens as specified in 3.11.2.

3.11.6 Pass/fail criteria. For each test specimen, the pass/fail criteria for this test will be that the anodic current density over the range of cupronickel potentials shall meet or exceed that of the control specimens tested in 3.11.2.

3.11.7 Seawater properties. Clean, natural seawater with sulfide concentrations less than 0.05 milligram per liter shall be used. Seawater properties during all portions of the test shall be in accordance with table II.

TABLE II. Seawater properties.

	Temperature (°C)	Salinity (parts per thousand)	pH	Dissolved Oxygen (mg/L)	Dissolved Oxygen (percent saturation)	Resistivity
MINIMUM	16	35	7.8	4	72	16
MAXIMUM	31	41	8.2	7	89	21

3.12 Identification and markings. Identification and markings shall be permanent and legible and shall include, as a minimum, the manufacturer's name, telephone number, and Commercial and Government Entity (CAGE) code in addition to labeling with instructions for use and appropriate hazard warnings. Products incompatible with cleaning of stainless steels shall be marked, "Do not use on stainless steels."

4. REGULATORY REQUIREMENTS. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4.1 Hazardous materials. The compound shall meet all applicable Federal codes and regulations concerning hazardous materials, including package marking.

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5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The compound shall have been sold previously to the Government or to the commercial market and meet the criteria in this CID, in order to be acceptable under this CID, or per the product listing in UIPI 5050-903.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor. A-A-59782 product PINs are constructed as follows:

a. Compound type:
   IH – Hydrochloric acid based
   IP – Phosphoric acid based
   OC – Citric acid based
   ON – Organic (non-citric) acid based
   OO – Other formulations, including acid-replacement or salt-based descaling technologies not otherwise listed.
b. Compound basis chemical percent: Two-digit numerical percentage by weight.
c. Package size: Up to three-digit size in gallons or liters, indicated by preceding G or L.
d. Material incompatibility: Optional suffix indicating material the product should not be used for, SS being stainless steels; other suffix codes as determined.

7.1.1 PIN examples.

a. A hydrochloric acid-based cleaning compound of 10 percent HCl, supplied in a 55-gallon drum would have a PIN of AA59782-IH10G55SS. Note the SS suffix is due to HCl being incompatible with stainless steels.
b. An organic acid-based cleaning compound of 15 percent carbolic acid blend, supplied in a 20-liter container would have a PIN of AA59782-ON15L20.

7.2 Source of documents.

7.2.1 ASTM International. ASTM standards are available online at www.astm.org.

7.2.2 Code of Federal Regulations (CFR). The CFR is available online at www.ecfr.gov.

7.2.3 SAE International. SAE publications are available online at www.sae.org.

7.2.4 Uniform Industrial Process Instructions (UIPI). UIPIs may be obtained online at https://www.navsea.navy.mil/Home/Shipyards/PHNS-IMF/.

7.3 CAS. CAS numbers are available online at www.cas.org.

7.4 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.
b. Quantity required.
c. When approvals are to be provided (see 3.7).
d. When SDSs are to be prepared (see 3.7.2).
e. Product conformance provisions.
f. Packaging requirements.

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7.5 National stock numbers (NSN). Known NSNs for heat exchanger cleaning compounds:

a. Hydrochloric acid based: 6850-01-465-9521, 6850-01-465-9246, 6850-01-457-8789, 6850-01-457-8790, 6850-01-549-3959, 6850-01-549-3960,6850-01-549-3964, 6850-01-509-0819, 6850-01-509-0820, and 6850-01-509-0815.

b. Phosphoric acid based: 6850-01-550-6520, 6850-01-550-6529, and 6850-01-550-6524.

c. Citric acid based: 6810-00-141-2942, 6850-66-164-0776, 6850-66-152-0080, 6850-66-152-0081, and 6850-66-168-5982.

d. Organic acid blend based: NSNs to be determined.

7.6 Procedures for use. Procedures for heat exchanger cleaning using these products may be found in:

a. (UIPI) 5050-903, Cleaning Marine Growth from a Component or System using Authorized Descaling Agent (ADA).

b. S9531-BH-MMI-010, Heat Exchanger Repair of Heat Exchangers, Coolers, and Distilling Plants.

7.7 Key words.

Acid cleaning
Acid replacement
Alternative descaling technology
Biobased
Calcium scale
Citric
Condenser
Evaporator
Hydrochloric
Inorganic
Lime scale
Low corrosion
Non-hazmat
Organic
Phosphoric
Salt based
Scale removal
Seawater system

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CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army – AT
Navy – SH
Air Force – 03

Review activities:
Army – EA, MR
Navy – AS
Air Force – 50, 68
DLA – GS, GS3
MISC – MP

Preparing activity:
Navy – SH
(Project 6850-2026-003)

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